

1 · PROJECT

Battery energy storage site, Sugar Land, TX (ERCOT): 375 kW / 1,221 kWh — three CATL EnerOne+ outdoor cabinets (407 kWh LFP each, liquid-cooled) with a Kehua Cabinet-375 PCS (integrated 30 kVA 480→380Y/220 V aux transformer). The package under review is ONE combined EMS + MBMU + fire control enclosure (NEMA 4, ~72×48×24 in) that consolidates two panels previously approved for the owner's Carson, CA site, redesigned for this application. Design revision K.

2 · ENGAGEMENT — WHAT WE ARE ASKING OF YOU

- **Independent design review.** Review the Rev K package (drawings, BOM, schedules, wire spec, load/battery calcs) against NEC 2023, NFPA 72, NFPA 855, and UL 508A. Direct any changes; our team turns revisions within 48 h.
- **Deficiency list & disposition.** Issue written comments; we resolve and resubmit until you are satisfied.
- **PE stamp at IFC.** Seal the final drawing set for construction once open items close. Nothing in the package is represented as engineer-certified today; internal design-QA checks are aids, not a certification.
- **AHJ interface.** Support the NFPA 72 secondary-power ruling (item 4.2) with the Fort Bend County / Sugar Land AHJ.
- **Vendor RFI review.** Review the two outstanding vendor RFIs (Kehua; CATL/Potter) when answered — both gate fabrication.

3 · WHAT YOU RECEIVE (ALL ON THE PORTAL)

Drawings: one-page ARCH E + 9-sheet A3 schematic set (PDF), with native KiCad and Altium sources. Documents: BOM (65 lines, proposed part numbers), fuse/breaker/terminal/label/cable schedules, 24 V load census, NFPA 72 battery worksheet, wire specification, 3-page system flowchart (fastest orientation), the original owner SOW, and the as-approved Carson baseline drawings. Interactive 3D site model and animated one-line simulator. Every drawing format is machine-verified against a single 174-net design database (BOM reconciled bidirectionally; KiCad and Altium proven electrically identical); the verification scripts ship in the package so you can re-run them.

PORTAL ACCESS

<https://smarttec.dev/z1power-sugarland-bess> · password provided in the covering e-mail

Password-controlled and unlisted. Please do not redistribute the link, password, or documents.

4 · OPEN ITEMS REQUIRING YOUR ENGINEERING JUDGMENT

- **4.1 SCCR.** Short-circuit current rating determination — utility available fault current is being requested; UL 508A Supplement SB4 worksheet to be completed with the panel shop.
- **4.2 NFPA 72 secondary power.** Battery manufacturer literature requires '24 h + 2 h sound and light'; whether the 2 h term applies to the panel FACP notification circuit is an AHJ interpretation. 8 Ah installed passes a 5-minute term, fails 2 h (13.8 Ah).
- **4.3 Fire releasing architecture.** Option B selected by owner: FACP (Potter ARC-100) → PSN-1000 releasing PSU → lockable supervised disconnect DS1 → cabinet aerosol initiators via connector JX1 pins 3/4. Aerosol pin functions and initiator ratings pend vendor RFI-002; the hardwired PCS trip (relay 26/27 → K1 → Kehua EPO, fail-safe/normally-energized) pends contact-sense confirmation in RFI-001.
- **4.4 Aux feeder tap.** NEC 240.21(B)(2) tap from the Kehua aux transformer breaker to panel OCPD Q2 — confirm basis; >30 A vendor breaker forces a feeder upsize from AWG 10.
- **4.5 Heat balance / A/C.** Enclosure cooling sized for sealed NEMA 4 + A/C; final size pends the site shading decision. FACP 49 °C ambient limit governs; a spot-cooling recommendation (G2) is documented.
- **4.6 Metering deviation (G1).** Owner SOW lists an EG4130 with 18 CTs; design carries one EG4115 with 9 CTs — owner approval in writing, or the metering section is redrawn.
- **4.7 Grounding.** Single N-G bond at the 120 V CPT (NEC 250.30); aux-system bond location inside the Kehua pends RFI-001.

5 · RESPONSIBILITY SPLIT

- **You (PE of record):** Engineering review, direction, AHJ interface, and the stamp. You control the standard of care; nothing is released for construction without your seal.
- **z1Power / SmartTec:** All drafting and revision labor, design database upkeep, vendor RFI follow-up, verification re-runs, panel-shop and site coordination. We revise; you approve.
- **Owner (STR Medical):** Decisions on documented deviations (G1), utility data, site information.

6 · SCHEDULE & COMMERCIAL

Requested: comment round within 2–3 weeks of access; stamp at IFC once open items close (vendor RFI answers expected within 10 business days of issue). Fee structure, contract form, and insurance/limits to be agreed by separate engagement letter — propose what works for your practice on our kickoff call.

7 · CONTACT

Yasir Jahangir — z1Power / SmartTec · yasir@futonix.com. Referred introduction via Umair. Questions on any drawing convention or the verification tooling: same address, same-day response.